

DESCRIPTION

The PT2033 is an audio processor designed for versatile application, it includes 3 stereo input selector with internal gain, master volume control with low frequency loudness compensation, individual channel output volume adjust and tone control.

If a system wants subwoofer signal output, traditional solution needs a lot of discrete components and OPAMP to consist of low pass filter; in the PT2033, just needs 2 capacitors only. The PT2033 combine the low pass filter inside the audio processor and user may determine low pass corner frequency by the capacitance. Broader supply range emphasis support to every application.

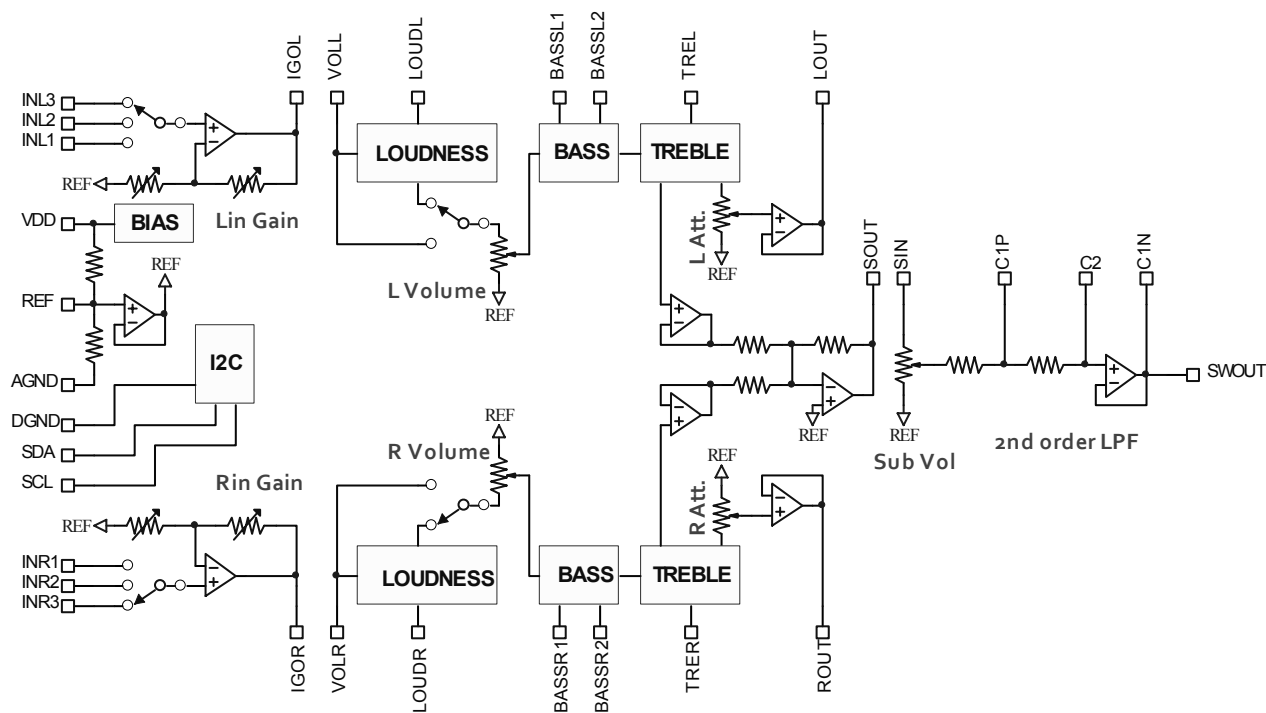
APPLICATIONS

- TV
- Mini compo
- Multimedia speaker
- Audio docking system

FEATURES

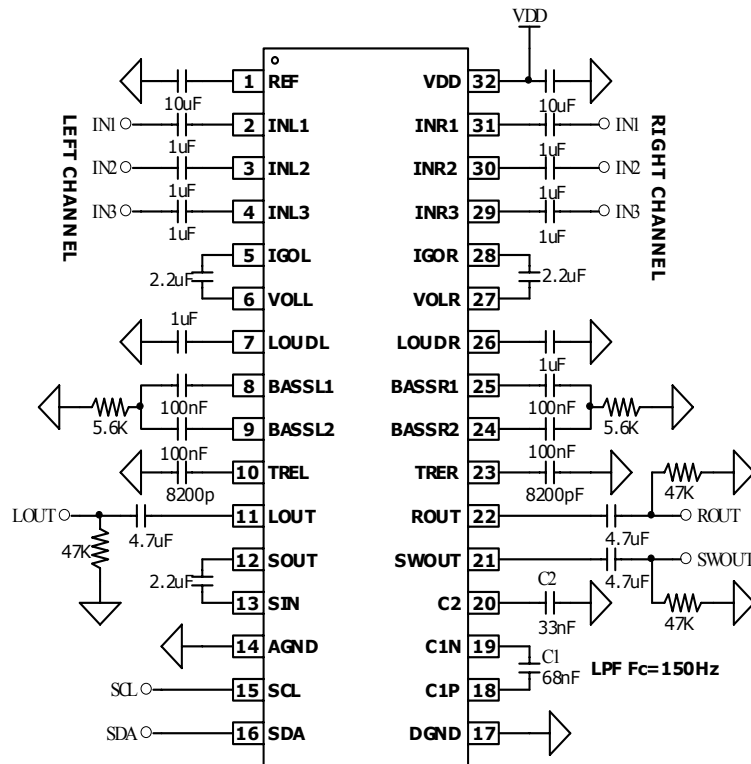
- CMOS technology
- Controlled by I²C interface
- 3 Stereo Inputs with gain selection.
- Input gain range, from 0dB to +11.25dB, 3.75dB per step
- 2 channels master volume: from 0 dB to -78.75dB, 1.25dB per step
- Built In channel mixer and lowpass filter for subwoofer output
- Subwoofer output volume : from 0dB to -37.5dB, 1.25dB per step
- Tone control (Bass and Treble): from -14dB to +14dB, 2dB per step
- Wide operation range(VDD from 4V to 10V)
- Low harmonic distortion (0.002%, Vo=200mVrms)
- Low noise and DC offset
- 32-pin SOP package

BLOCK DIAGRAM

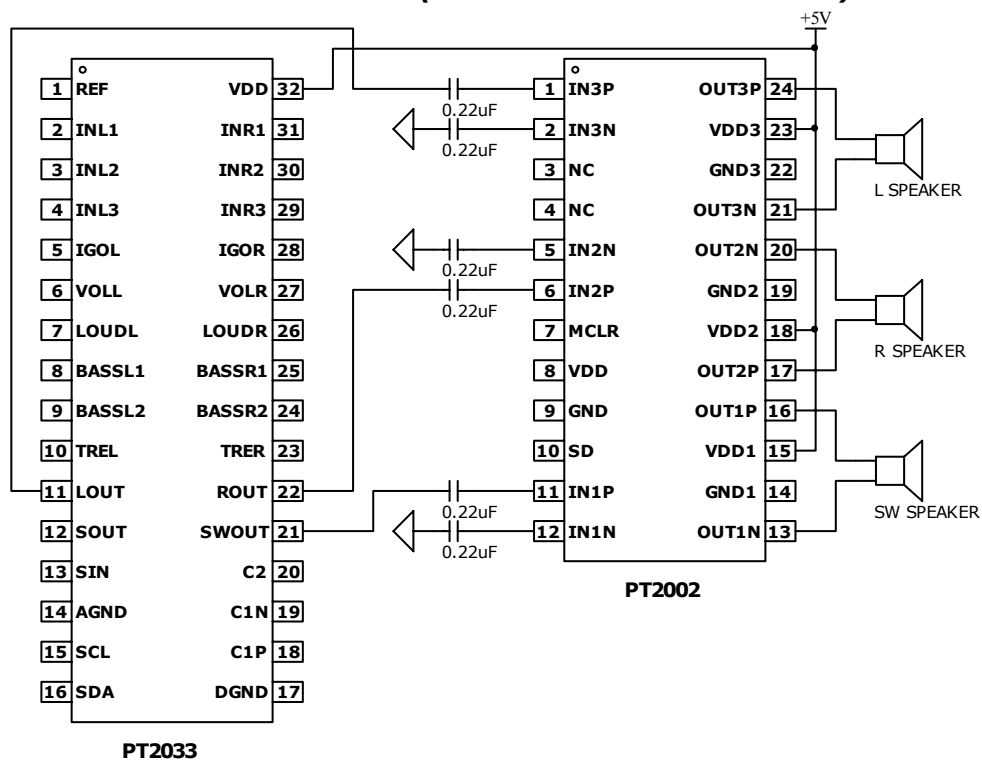


APPLICATION CIRCUIT

STANDARD APPLICATION CIRCUIT



CHANNEL SPEAKER SYSTEM (WIRING SIMPLIFIED)

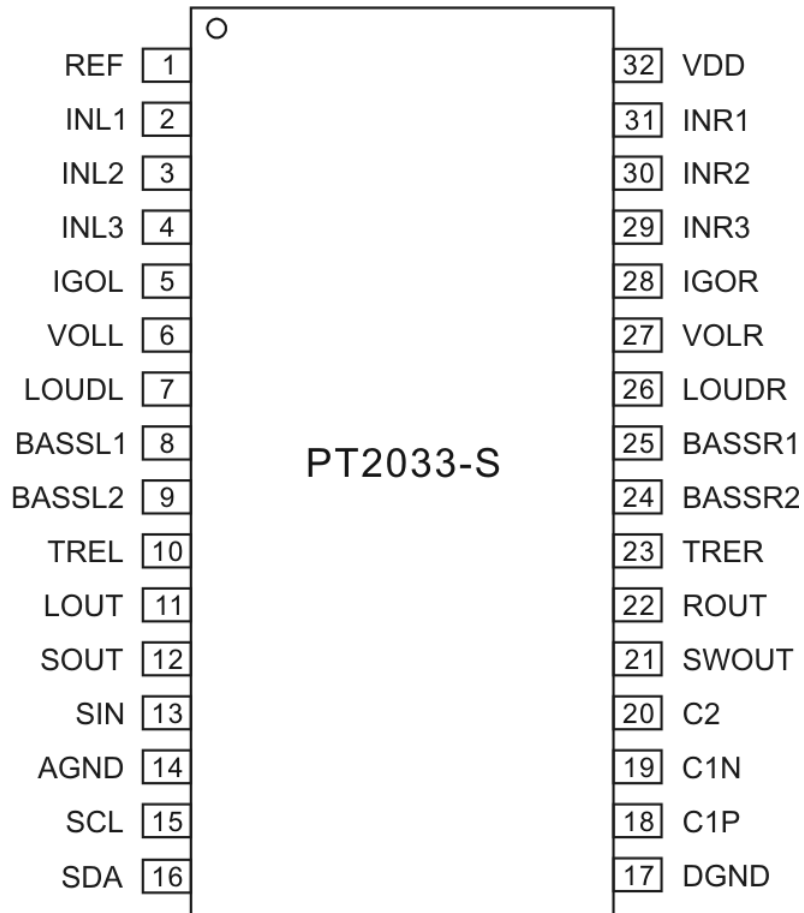




ORDER INFORMATION

Valid Part Number	Package Type	Top Code
PT2033-S	32 Pins, SOP, 300mil	PT2033-S

PIN CONFIGURATION





PIN DESCRIPTION

Pin Name	I/O	Description	Pin No.
REF	-	Analog reference voltage (1/2 VDD)	1
INL1	I	Left channel input 1	2
INL2	I	Left channel input 2	3
INL3	I	Left channel input 3	4
IGOL	O	Left channel buffer output	5
VOLL	I	Left channel volume controller input	6
LOUDL	I	Capacitor node for left channel loudness	7
BASSL1	I	Capacitor node 1 st for left channel bass filter	8
BASSL2	O	Capacitor node 2 nd for left channel bass filter	9
TREL	I	Capacitor node for left channel treble filter	10
LOUT	O	Left channel output	11
SOUT	O	L+R mixer (mono) output	12
SIN	I	Low pass filter input	13
AGND	I	Analog ground	14
SCL	I	I ² C clock input	15
SDA	I	I ² C data input	16
DGND	-	Digital ground, connect with AGND in shortest route.	17
C1P	I	Node P for low pass filter ext cap	18
C1N	I	Node N for low pass filter ext cap	19
C2	I	2 nd node for low pass filter ext cap	20
SWOUT	O	Subwoofer output	21
OUTR	O	Right channel output	22
TRER	I	Capacitor node for right channel treble filter	23
BASSR2	O	Capacitor node 2 nd for right channel bass filter	24
BASSR1	I	Capacitor node 1 st for right channel bass filter	25
LOUDR	I	Capacitor node for right channel loudness	26
VOLR	I	Right channel volume controller input	27
IGOR	O	Right channel buffer output	28
INR3	I	Right channel input 3	29
INR2	I	Right channel input 2	30
INR1	I	Right channel input 1	31
VDD	-	Power supply	32

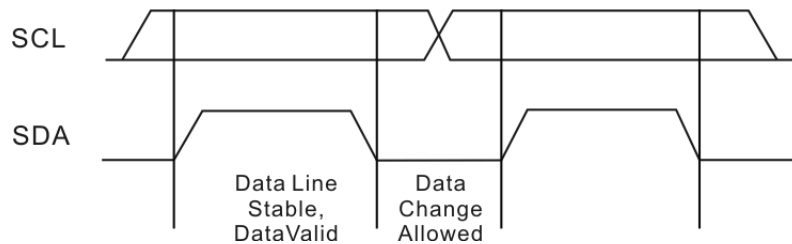
FUNCTIONAL DESCRIPTION

BUS INTERFACE

All functions of the PT2033 are controlled by the I²C interface, the interface is consisting by SDA and SCL pins. Detail protocol of the I²C bus will discuss on the next section. It should be noted that the bus level pull-up resistors connected to the PT2033 positive supply voltage may required in some application especially the MCU output high level is no enough.

DATA VALIDITY

A data on the SDA Line is considered valid and stable only when the SCL Signal is in HIGH State. The HIGH and LOW State of the SDA Line can only change when the SCL signal is LOW. Please refer to the figure below.



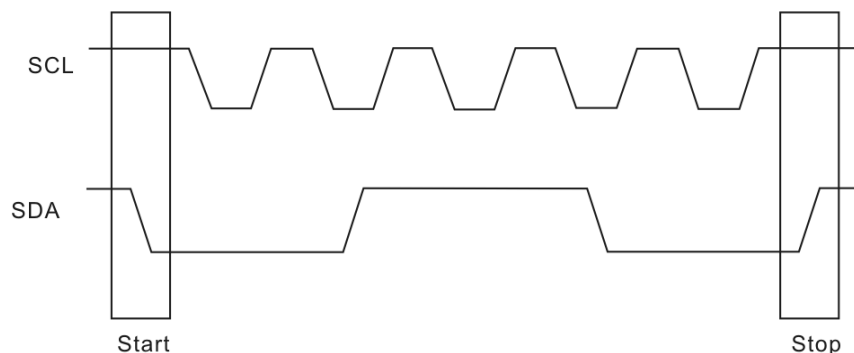
START AND STOP CONDITIONS

A Start Condition is activated when

- 1) The SCL is set to HIGH and
- 2) SDA shifts from HIGH to LOW State.

The Stop Condition is activated when

- 1) SCL is set to HIGH and
- 2) SDA shifts from LOW to HIGH State. Please refer to the timing diagram below.

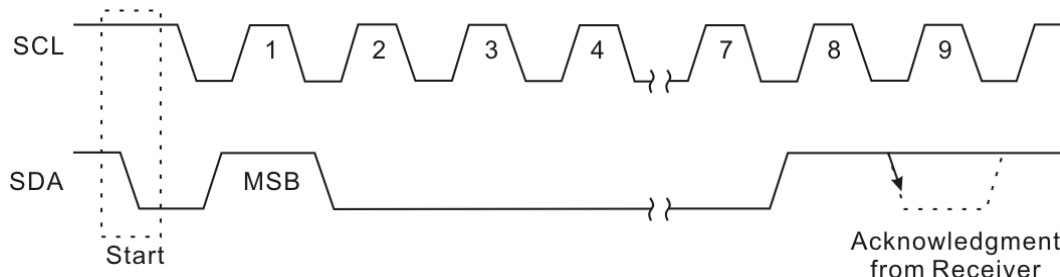


BYTE FORMAT

Every byte transmitted to the SDA Line consists of 8 bits. Each byte must be followed by an Acknowledge Bit. The MSB is transmitted first.

ACKNOWLEDGE

During the Acknowledge clock pulse, the SDA output port of the master device (μP) would be sets on Hi-Z state, the peripheral (ex: audio processor) that acknowledges has to pull-down (Low) the SDA line during the Acknowledge Clock Pulse so that the SDA Line is in a Stable Low State during this Clock Pulse. Please refer to the diagram below.



The slave device that has been addressed to generate an Acknowledge after receiving each byte, otherwise, the SDA Line will remain at the High level in period of the ninth (9th) clock pulse. In this case, the host controller will generate a STOP sign in order to abort the transfer mission.

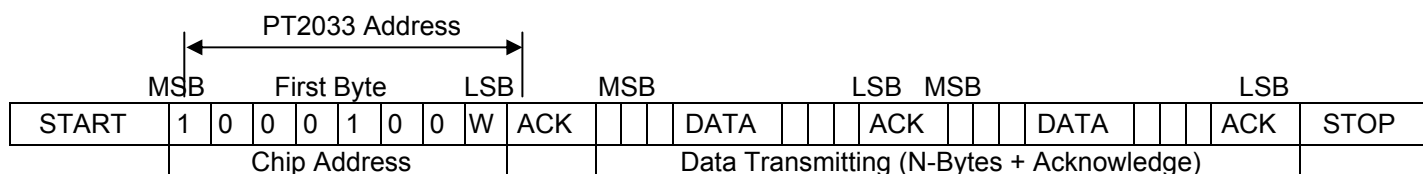
TRANSMISSION WITHOUT ACKNOWLEDGE

If the application does not need to verify the Acknowledge signal that generated by the slave device is right or not, host controller can just bypass the acknowledge check and transmit next data byte to the slave device. If this approach is used, there are greater chances of faulty operation as well as decrease in noise immunity.

INTERFACE PROTOCOL

The interface protocol sequence was defined in below section:

- A Start sign
- A Chip Address of the desire slave device. The W Bit must be "0" (written). The PT2033 will always response an Acknowledge on the end of each byte.
- A Data Sequence (N-Bytes + Acknowledge)
- A Stop Condition



Notes:

ACK=Acknowledge

Recommend SCL Frequency=100KBits/S



SOFTWARE SPECIFICATION

PT2033 CHIP ADDRESS

The PT2033 chip address is 0x88 AND binary table is shown on below.

MSB						LSB	
1	0	0	0	1	0	0	0

DATA BYTES

MSB							LSB	Function
0	0	B2	B1	B0	A2	A1	A0	Volume control
1	1	0	B1	B0	A2	A1	A0	Speaker ATT L
1	1	1	B1	B0	A2	A1	A0	Speaker ATT R
1	0	0	B1	B0	A2	A1	A0	Speaker ATT subwoofer
0	1	0	G1	G0	S2	S1	S0	Audio switch
0	1	1	0	C3	C2	C1	C0	Bass control
0	1	1	1	C3	C2	C1	C0	Treble control

where Ax=1.25dB/steps; Bx=10dB/steps; Cx=2dB/steps; Gx=3.75dB/steps

VOLUME

The table below gives a detailed description of the Volume Data Bytes. For example, a volume of -37.5dB is given by 0 0 1 1 1 1 0.

MSB							LSB	Function
0	0	B2	B1	B0	A2	A1	A0	Volume 1.25dB steps
					0	0	0	0
					0	0	1	-1.25
					0	1	0	-2.5
					0	1	1	-3.75
					1	0	0	-5
					1	0	1	-6.25
					1	1	0	-7.5
					1	1	1	-8.75
0	0	B2	B1	B0	A2	A1	A0	Volume 10dB steps
		0	0	0				0
		0	0	1				-10
		0	1	0				-20
		0	1	1				-30
		1	0	0				-40
		1	0	1				-50
		1	1	0				-60
		1	1	1				-70



SPEAKER ATTENUATORS

The table below gives a detailed description of the speaker attenuators data bytes. For example, an attenuation of 30dB on the Speaker R (Right) is given by: 1 1 1 1 1 0 0 0.

MSB							LSB	Function
1	1	0	B1	B0	A2	A1	A0	Speaker L
1	1	1	B1	B0	A2	A1	A0	Speaker R
1	0	0	+B1	B0	A2	A1	A0	Speaker ATT subwoofer
					0	0	0	0
					0	0	1	-1.25
					0	1	0	-2.5
					0	1	1	-3.75
					1	0	0	-5
					1	0	1	-6.25
					1	1	0	-7.5
					1	1	1	-8.75
			0	0				0
			0	1				-10
			1	0				-20
			1	1				-30
			1	1	1	1	1	Mute

AUDIO SWITCH DATA BYTE

The following table shows the detailed description of the Audio Switch Data Bytes. For example, a Stereo 1 Input with Gain of +11.25dB Loudness ON is given by: 0 1 0 0 0 0 0 0.

MSB							LSB	Function
0	1	0	G1	G0	S2	S1	S0	Audio switch
						0	0	Stereo 1
						0	1	Stereo 2
						1	0	Stereo 3
						1	1	Stereo 4
					0			Loudness ON
					1			Loudness OFF
			0	0				+11.25dB
			0	1				+7.5dB
			1	0				+3.75dB
			1	1				0dB



BASS AND TREBLE DATA BYTES

The following table shows a detailed description of the Bass and Treble Data Byte. For example a Treble at -12dB is given by: 0 1 1 1 0 0 0 1 (0x71).

MSB							LSB	Function
0	1	1	0	C3	C2	C1	C0	Bass
0	1	1	1	C3	C2	C1	C0	Treble
				0	0	0	0	-14
				0	0	0	1	-12
				0	0	1	0	-10
				0	0	1	1	-8
				0	1	0	0	-6
				0	1	0	1	-4
				0	1	1	0	-2
				0	1	1	1	0
				1	1	1	1	0
				1	1	1	0	2
				1	1	0	1	4
				1	1	0	0	6
				1	0	1	1	8
				1	0	1	0	10
				1	0	0	1	12
				1	0	0	0	14

Unit: dB

MIXER OUTPUT

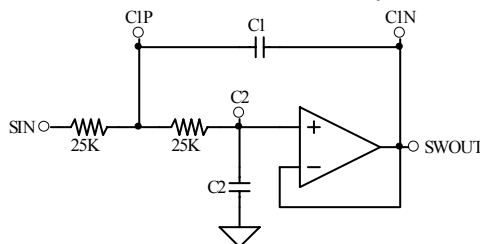
The mono mixer output (SOUT) of the PT2033 is mix left and right channel signal together, and it's output gain is fixed on -6dB to prevent clipped, this situation may happen on both channels are working on maximum output level. Mixer output next will feed into low pass filter for further signal processing.

SUBWOOFER LOW PASS FILTER

The main channel stereo signal in front of output volume stage is feed into a stereo-to-mono mixer; this stage has a fixed -6dB voltage gain to prevent output clipped. After a DC block couple capacitor the signal into the lowpass stage, this stage have internal resistor and OPAMP buffer, certainly subwoofer volume control also included. The low pass filter is formed by a Sallen Key type, due to the internal series resistor is fixed on 25KΩ so corner frequency determined by the external capacitors, C1 and C2.

$$\frac{V_o}{V_i} = \frac{1}{S^2 C_1 C_2 R_1 R_2 + S C_2 (R_1 + R_2) + 1} = \frac{\frac{1}{C_1 C_2 R_1 R_2}}{S^2 + S \frac{1}{C_1 (R_1 // R_2)} + \frac{1}{C_1 C_2 R_1 R_2}}$$

This is the transfer function of a lowpass filter, and below is the real implement circuit.



In a 3 pieces speaker system, the main speaker is handling most of audio range except extreme low frequency; it is because deep bass sound needs very large diameter subwoofer to generate, and in most of case the system only has one subwoofer in there because considering about price and easy to placed. Basically extreme low frequency audio energy radiation pattern is omni-direction type, therefore, just use one subwoofer and setting on a very low corner frequency is a sufficient a cost effective audio system. In most case the subwoofer LF corner frequency should below 300Hz to prevent audible direction steering, in the electric characteristics chapter shows recommend crossover frequencies and respective external capacitors.



ABSOLUTE MAXIMUM RATINGS

Parameter	Symbol	Rating	Unit
Operating supply voltage	Vs	10.5	V
Operating temperature	Topr	-40 to +85	°C
Storage temperature	Tstg	-65 to +150	°C

QUICK REFERENCE DATA

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply voltage	Vs	4	9	10	V
Max. input signal handling	VCL	2.5	2.7	-	Vrms
Total harmonic distortion (1Vrms, 1KHz)	THD	-	0.003	0.01	%
Signal to noise ratio	S/N	-	105	-	dB
Channel separation (f=1KHz)	Sc	-	90	-	dB
Volume control 1.25dB step	-	-78.75	-	0	dB
Bass & treble control 2dB step	-	-14	-	+14	dB
Balance control 1.25dB step	-	-37.5	-	0	dB
Subwoofer control 1.25dB step	-	-37.5	-	0	dB
Input gain 3.75dB step	-	0	-	11.25	dB
Mute attenuation	-	-	100	-	dB



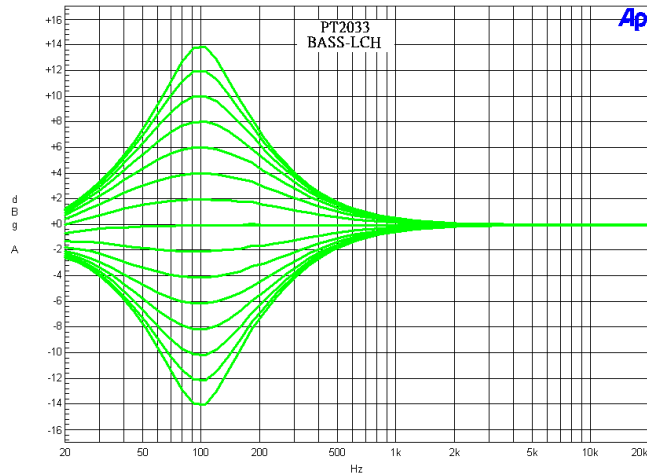
ELECTRICAL CHARACTERISTICS

Unless otherwise specified: Ta=25°C, VDD=9V, RL=100KΩ, Rg=600Ω, all controls flat (G=0), f=1KHz

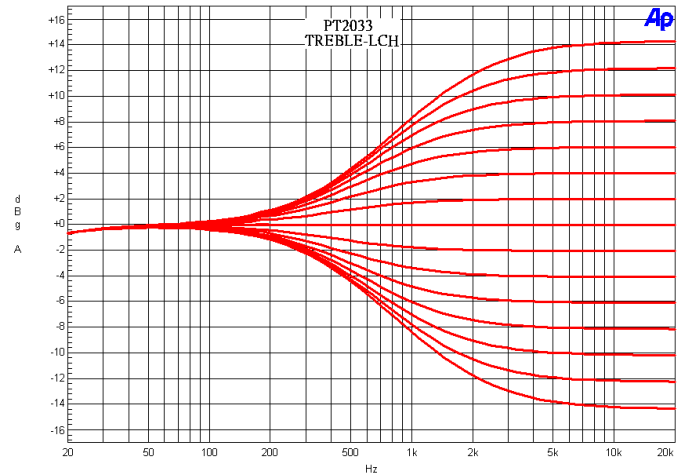
Parameter	Symbol	Test Condition	Min.	Typ.	Max.	Unit
Supply						
Supply voltage	VDD	-	4	9	9.7	V
Supply current	Is	-	18	25	33	mA
Input Selectors						
Input resistance	R _{II}	Input 1, 2, 3	35	50	65	KΩ
Clipping level	V _{CL}	Vol=-8.75dB; THD=0.3%	2.5	2.7	-	Vrms
Input separation	S _{IN}	-	80	100	-	dB
Min. input gain	G _{INmin}	-	-1	0	1	dB
Max. input gain	G _{INmax}	-	10.5	11.25	12	dB
Step resolution	G _{Inst}	-		3.75		dB
Gain set error	E _A	-	-1	0	1	dB
Minimum load	RL	IGOL, IGOR	10	-	-	KΩ
Volume Control						
Min. attenuation	A _{VMIN}	-	-1	0	1	dB
Max. attenuation	A _{VMAX}	-	-75	-78.75	-82	dB
Step resolution	A _{STEP}	-	1.15	1.25	1.3	dB
Attenuation set error	E _A	Vol=0 to -60dB	-1	0	1	dB
Speaker Attenuators (LOUT, ROUT, SWOUT)						
Control range	C _{RANGE}	-	35	37.5	40	dB
Step resolution	S _{STEP}	-	1.15	1.25	1.35	dB
Attenuation set error	E _A	-	-1	0	1	dB
Output mute attenuation	A _{MUTE}	-	100	110	-	dB
Bass Control (1)						
Control range	G _b	Max. Boost/Cut	±12	±14	±16	dB
Step resolution	B _{STEP}	-	1.7	2	2.3	dB
Feedback resistance	R _B	-	37	47	57	KΩ
Treble Control (1)						
Control range	G _t	Max. Boost/Cut	±12	±14	±16	dB
Step resolution	T _{STEP}	-	1.7	2	2.3	dB
Feedback resistance	R _T	-	-	-	-	KΩ
Mono Mixer						
Output gain	M _g	Single channel input only	-5	-6	-7	dB
		L=R, both channels input	-1	0	+1	
Low Pass Filter						
Internal filter resistance	R _{int}		20	25	30	KΩ
Audio Outputs (LOUT, ROUT, SWOUT)						
Clipping level	V _{OCL}	THD=0.3%	2.5	2.7	-	Vrms
DC voltage level	V _{OUT}	-	0.49	0.5	0.51	VDD
Minimum load	RL	-	5	-	-	KΩ
General						
Signal to noise ratio	SNR	All Gain=0dB, A-weighted	-100	-105	-	dBV
		All Gains=0dB, Muted	-105	-110	-	
Distortion	THD	AV=0, Vin=1Vrms	-	0.007	0.01	%
		AV=0, Vin=200Vrms	-	0.003	0.008	
Channel separation	C _s	-	85	95	-	dB
Ripple rejection	PSRR	Vr=200mVrms F=200Hz	60	72	-	dB
I ² C Bus						
Input low voltage	V _{IL}	-	-	-	1	V
Input high voltage	V _{IH}	-	3	-	-	V
Input current	I _{IN}	-	-5	-	+5	μA

Notes:

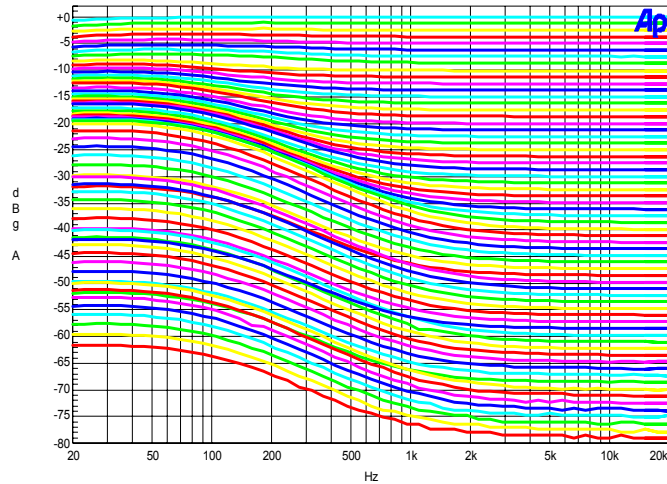
- For the Bass and Treble Response, please, refer to the diagram below. The center frequency and quality of the resonance behavior can be selected by the external circuitry. A standard first order bass response can realize by a standard feedback network.
- The selected input is grounded thru the 2.2μF capacitor.



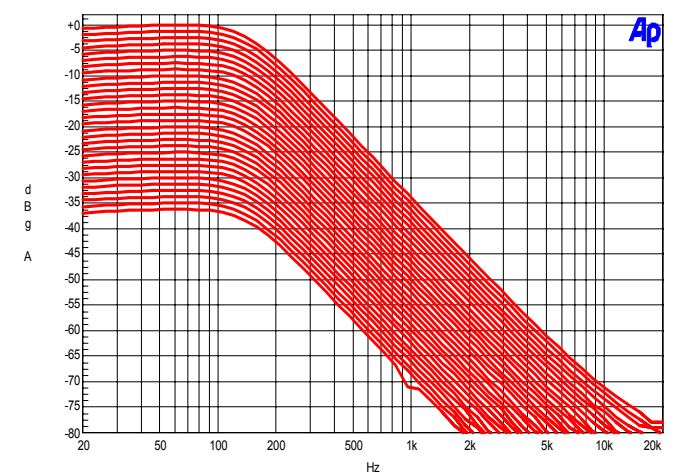
Bass control characteristics



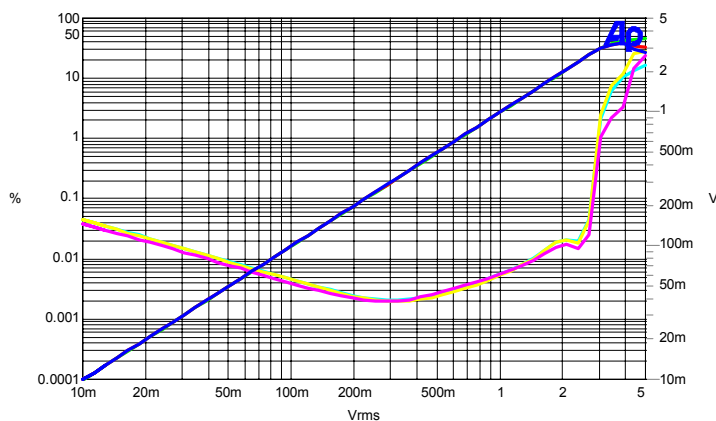
Treble control characteristics



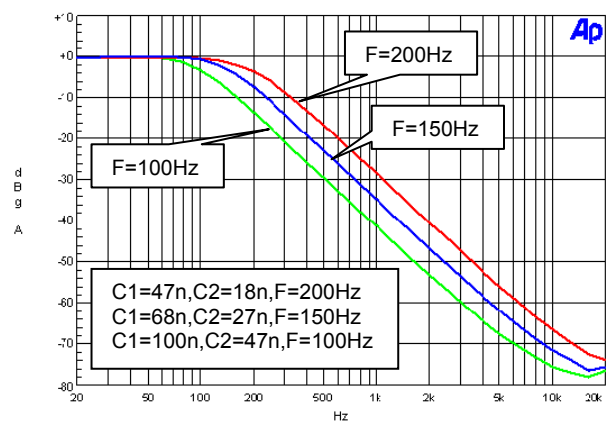
Loudness characteristics vs. Volume



Low pass filter characteristics with output level attenuation



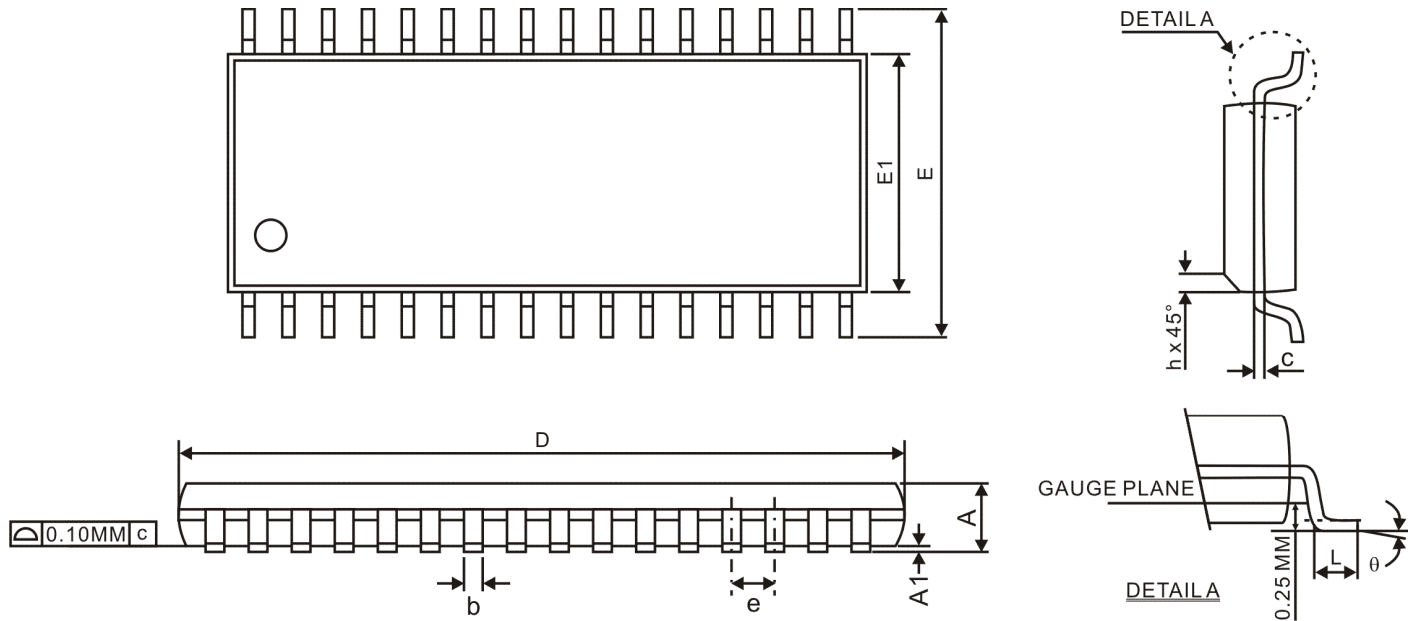
THD, output voltage vs. Input voltage



Low pass filter crossover characteristics with different external capacitor

PACKAGE INFORMATION

32 PINS, SOP, 300MIL



Symbol	Min.	Typ.	Max.
A	-	-	2.65
A1	0.10	-	-
b	0.31	-	0.51
c	0.20	-	0.33
e	1.27 BSC.		
D	20.32	-	20.73
E	10.00	-	10.65
E1	7.40	-	7.60
L	0.38	-	1.27
θ	0°	-	8°

Notes:

1. Refer to JEDEC MO-119 AC
2. Unit: mm



IMPORTANT NOTICE

Princeton Technology Corporation (PTC) reserves the right to make corrections, modifications, enhancements, improvements, and other changes to its products and to discontinue any product without notice at any time.

PTC cannot assume responsibility for use of any circuitry other than circuitry entirely embodied in a PTC product. No circuit patent licenses are implied.

Princeton Technology Corp.
2F, 233-1, Baociao Road,
Sindian, Taipei 23145, Taiwan
Tel: 886-2-66296288
Fax: 886-2-29174598
<http://www.princeton.com.tw>